Intern-S2-Mobius: Foundation Model with Decoupled Knowledge and Reasoning

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We introduce ***Mobius-v0***, an architecture that endogenously supports **Backward Residual Connection** and **Dynamic Latent Reasoning** by disentangling knowledge vectors and reasoning operators within the base model. Mobius comprises a globally shared Memory that stores knowledge vectors and multiple Reasoners that iteratively achieve compositional reasoning. Using hidden states as cache and carrier, reasoners repeatedly query memory for required knowledge-vectors, while the knowledge is transmitted back to reasoning operators. Through this knowledge-reasoning-separation architecture, Mobius achieves better knowledge compression and reasoning efficiency. Built upon Mobius-v0 architecture: 1) Our trained-from-scratch 7B model achieves similar downstream score as a 7B Transformer baseline with 62.6% of baseline's training data. 2) Our **Intern-S2-Mobius**, continually-pretrained from Qwen3.5-35B, achieves similar downstream score while delivering nearly 4× end-to-end inference speedup.

1. The development bottleneck of the Foundation Models

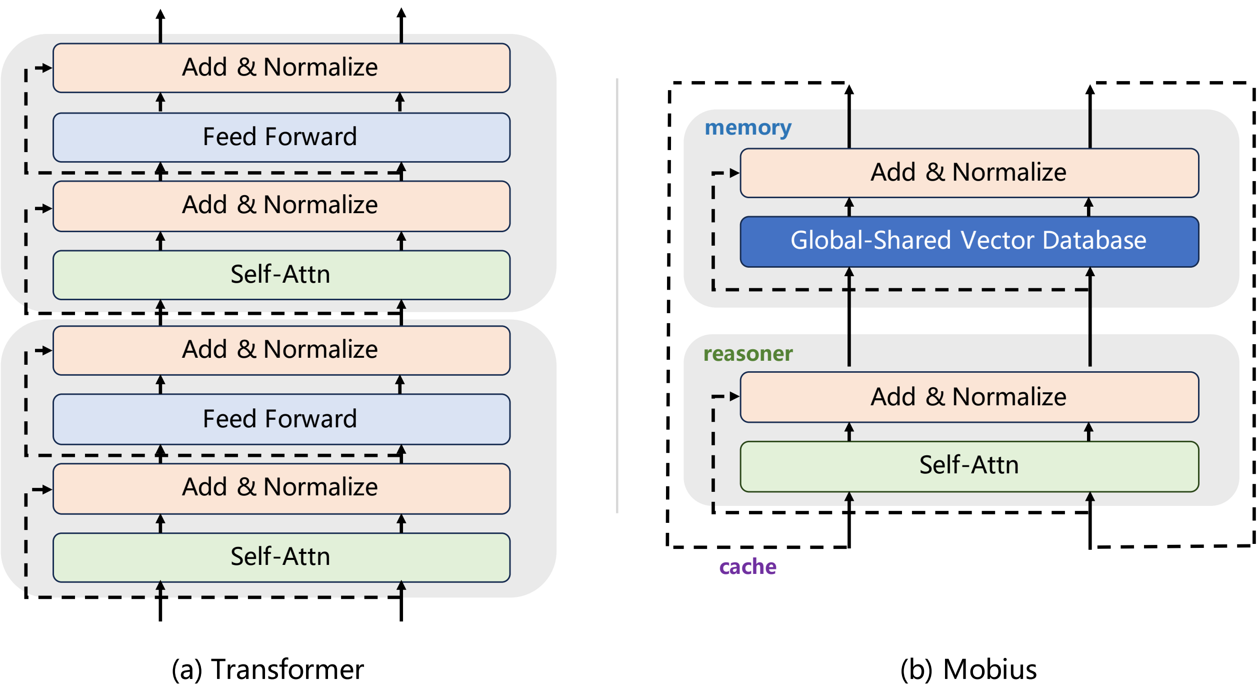

Figure 1: The Comparison between Transformer and Mobius.

The Transformer [69] stands as the most prevalent and powerful architectural paradigm to date, permeating virtually every domain—from language and vision, to video and scientific computing [4, 15, 16, 81]. Along its evolutionary trajectory, two pivotal research directions have emerged. The first preserves the model architecture intact while increasing expenditure elsewhere to enhance capability: scaling model parameters, expanding training corpora, and elongating reasoning chains typically endow models with richer knowledge and the

∗ Model is available at https://huggingface.co/internlm/Intern-S2-Mobius capacity to tackle more intricate problems [33, 37, 45, 71]. The second reduces training and inference overhead by lowering architectural complexity to improve practicality: motivated by the widely held view that the quadratic complexity of self-attention constitutes a bottleneck for ultra-long contexts, linear attention mechanisms such as SSM and GDN have been proposed [10, 25, 38, 78], and their variants have since been integrated into mainstream contemporary architectures.

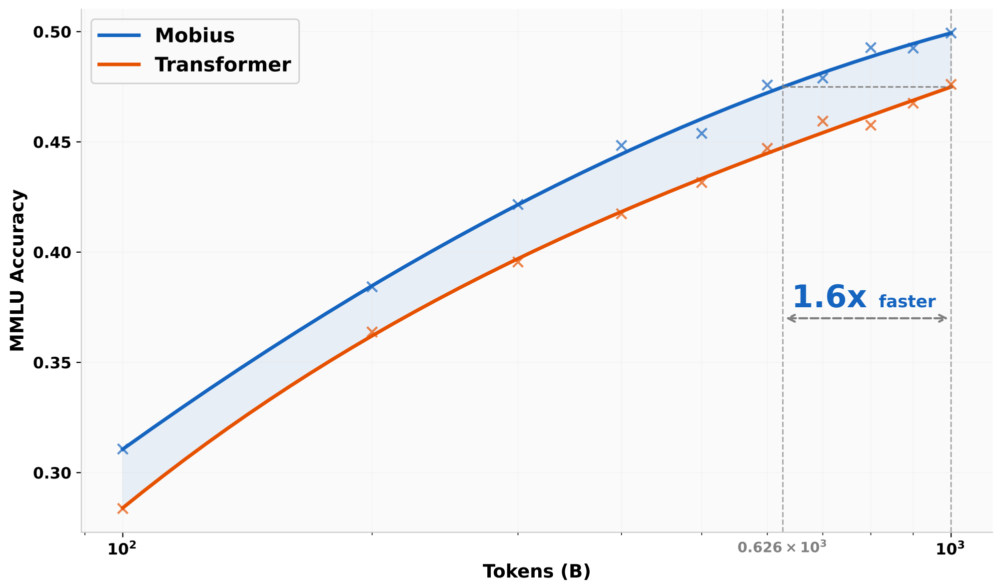

Figure 2: The MMLU score of Mobius and Transformer pre-trained from scratch.

Yet both approaches are now hitting their limits. On one hand, scaling data and parameters has yielded increasingly powerful models at high returns on investment, yet the Scaling Law has gradually plateaued and is approaching diminishing marginal returns [33, 37]. Long chains of thought have served us well in domains such as mathematics, code, and physics, yet models tend to produce verbose, tangential output regardless of problem difficulty—a trait typically regarded in human society as a sign of insufficient intelligence [49, 54]. On the other hand, reducing the computational complexity of full attention does bring certain efficiency gains, yet it has become increasingly apparent that the efficiency improvements achieved at the cost of sacrificing a substantial portion of model capability have not made models truly affordable enough for commercial deployment [3].

Confronted with this bottleneck in AI development, we pursue an alternative path: **increasing architectural complexity to raise the ceiling of model intelligence, thereby reducing the end-to-end overhead.** To this purpose, we propose the ***Mobius***, which decouples the binding between knowledge vectors and reasoning operators, thereby constructing a shared knowledge vector database accessible to all reasoning operator groups. Although the large scale of this shared vector database renders each inference activation highly sparse, introducing greater memory access pressure and lower per-pass forward efficiency, we find that this implementation achieves nearly 4$\times$ end-to-end inference speedup over the Transformer architecture with same parameters while maintaining equivalent reasoning accuracy.

The efficiency gains of Mobius reasoning stem primarily from two sources: a more flexible activation path during inference, and a more dynamic latent-space iteration, which together endow Mobius with a more efficient and concise output pattern. The first advantage arises because, unlike the combination of hierarchical storage and forward residual connections in Transformer, Mobius's shared storage natively introduces **Backward Residual Connection**. This mechanism means that not only can shallow hidden states access deep-layer knowledge, but deep hidden states can also access shallow-layer knowledge, which enhances the compositional generalization across different layers and thereby accelerates the synthesis of critical information. The second advantage arises because Mobius natively introduces **Dynamic Latent Reasoning**: whereas Transformer uses tokens as the medium of information transfer and requires traversing all layers to complete the inference of a single token, Mobius can iterate and refine latents against the full knowledge repository within just a few layers, rather than requiring multiple full-layer iterations as in traditional latent reasoning. These latents are not tightly bound to any specific token, and only at deeper layers are multiple tokens decoded synchronously. This approach both increases the density of information transfer and dynamically allocates varying computational costs to different tokens. Ultimately, Mobius completes the same reasoning task with markedly fewer high-quality tokens, achieving substantially higher end-to-end inference efficiency than Transformer.

Table 1: Performance comparison across general and scientific benchmarks. The higher score in each row is highlighted in **bold**.

General Tasks

Benchmark	Intern-S2-Mobius-35B	Qwen3.5-35B
MMLU Pro	**89.05**	85.31
GPQA Diamond	**80.81**	80.24
IMO Bench	**81.25**	77.50
AIME 2026	**95.31**	92.08
HMMT 2026	**85.51**	78.50
UGD hard	73.02	**78.02**
AMO	**58.00**	50.00
SimpleQA	**28.90**	21.39
HLE	19.11	**22.40**
AVG Score	**67.88**	65.05

Scientific Tasks

Benchmark	Intern-S2-Mobius-35B	Qwen3.5-35B
Biology-Instructions	**51.40**	3.77
Mol-Instructions	**45.73**	21.70
MolecularIQ	**59.29**	29.13
AVG Score	**52.14**	18.20

2. What inspired the design of Mobius?

In Transformer-based architectures, the Feed-Forward Network is conventionally regarded as responsible for knowledge storage [20, 52], Hidden States for information transmission, and Self-Attention for compositional reasoning [17, 69].

Concurrently, owing to the hierarchical structure of Transformer, Self-Attention at each layer primarily processes knowledge inputs received from the preceding layer and induces the FFN at the current layer to produce outputs required by the next layer. Although residual connections establish more flexible connections across layers [31, 34, 67], only shallow Hidden States can access deep-layer knowledge, while deep-layer Self-Attention can only process knowledge originating from shallow layers; the model remains incapable of information transfer in the opposite direction.

To construct information transfer in the reverse direction, current models predominantly rely on Chains of Thought [26, 71]. After each forward pass, the model generates a new token that serves to activate knowledge in the next inference step. Through iterative token generation, the model continuously extracts valuable knowledge from the FFN until this knowledge suffices to produce critical tokens or even the final answer.

The joint construction of artificial intelligence models using exclusively hierarchical structures, forward residual connections, and token-mediated chains of thought is inefficient. To equip models with sufficient knowledge for answering complex questions, an inordinate amount of computation is expended on producing lengthy, redundant reasoning chains. To mitigate this issue, we have redesigned the model architecture and propose Mobius.

2.1. Mobius' first innate talent —— Backward Residual Connection

Residual connections have become one of the most critical components in modern deep learning models [31]. Seemingly minor, they bear the crucial responsibility of transmitting inter-layer information during forward propagation and stabilizing gradient magnitudes during backpropagation. [32, 75] Although numerous variants

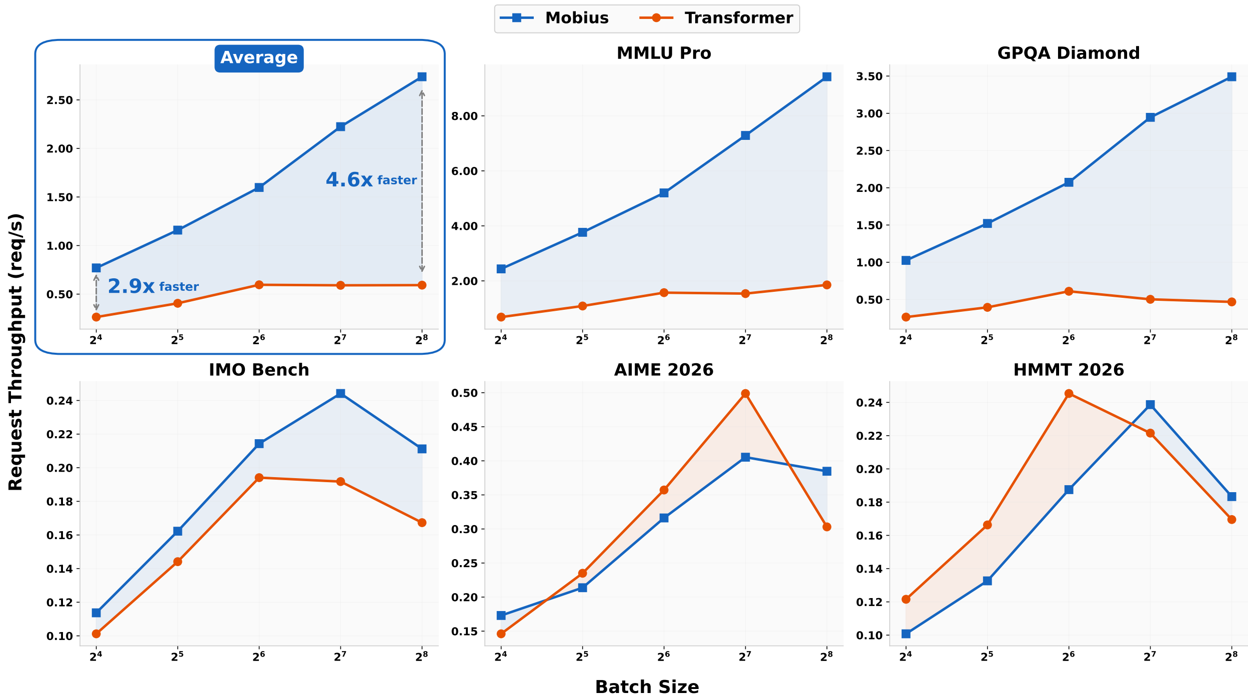

Figure 3: The inference efficiency of Mobius continually pre-trained from Qwen3.5.

of residual connections have since emerged, such as Hyper-Connection and Attention-Residual [39, 74, 80], all mainstream residual to date retain their unidirectional nature. For instance, during forward propagation, residual only convey shallow-layer information to deeper layers. [34] This unidirectionality entails significant drawbacks: if certain critical knowledge fails to be activated during shallow-layer computation, the model may struggle to decode valuable tokens in that reasoning round, and can only resort to generating low-information tokens via chains of thought to proceed to the next reasoning step.

To mitigate this phenomenon, rendering residual connections bidirectional is necessary—that is, enabling deep layers to access shallow-layer knowledge during forward propagation. However, a direct implementation of such backward residual connections is infrastructure-unfriendly, as it may introduce more complex computation graphs and harder-to-parallelize asynchrony. To this end, we opt for an indirect realization of backward residual connections: different layers share a single, oversized knowledge repository, granting every layer the opportunity to access all knowledge within the model. Empirical results ultimately corroborate that this form of backward residual connection contributes meaningfully to improving the model's end-to-end inference efficiency.

2.2. Mobius' second innate talent —— Dynamic Latent Reasoning

Long chains of thought (Long CoT) have become an indispensable component in large language model reasoning [26, 45, 71]. Empowered by this technique, contemporary large language models can now solve a wide spectrum of complex reasoning problems, spanning mathematics, physics, and code generation. Yet the cost of Long CoT is exorbitant. On one hand, longer reasoning chains entail the generation of more tokens, with generation cost growing non-linearly with chain length. On the other hand, current models frequently adopt a trial-and-error-and-correct approach when tackling complex problems, resulting in substantial token redundancy during extended reasoning. Moreover, such models tend to produce excessively verbose responses even for simple problems. This verbosity stems partly from the residual connections discussed above, and partly from the fact that the minimal unit of our current reasoning is the token—a discrete, low-information-density storage [9].

Mobius internalizes processes such as deliberation, trial-and-error, and refinement into the optimization of a continuous vector, and enables the model to dynamically allocate computational budget to different tokens. This

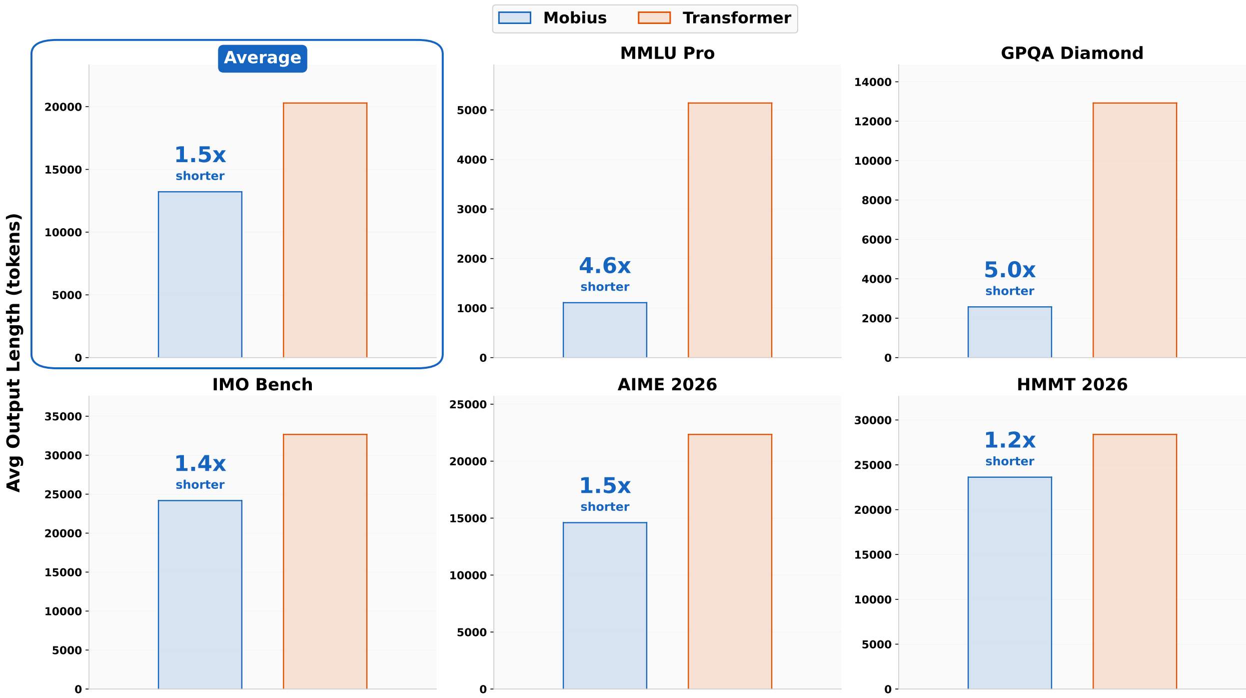

Figure 4: The average output length of Mobius continually pre-trained from Qwen3.5.

creates the opportunity for the model to produce fewer superfluous tokens while achieving more efficient and more intelligent reasoning. Mobius can be regarded as an upgraded synthesis of the Looped Transformer and the Diffusion Language Model [21, 43, 55], employing more efficient latent reasoning and parallel prediction. First, Mobius operates at a higher loop frequency, completing one representation iteration with extremely few layers. Second, during inference, Mobius acquires joint representations of multiple tokens simultaneously through more compact, high-information-density continuous vectors. This, on one hand, alleviates the pressure from parallelizing multiple hidden states, reduces the complexity of KV-cache management during the looping process, and enhances the continuous differentiability of the vector iteration process. Meanwhile, this native latent reasoning endows the model with a more dynamic and unfettered iteration process, without constraining the model to decode a fixed number of tokens at specific iteration steps.

2.3. One Stone Two Birds —— Disentangling Knowledge Vectors and Reasoning Operators

Overall, Mobius achieves these two innate talents by decoupling the knowledge storage module, the FFN, from its layer-wise binding, and constructing a globally shared knowledge-vector database. First, this grants all reasoning operators, the Self-Attention, read access to the entire body of knowledge. Further, the Self-Attention gains the opportunity to access all knowledge within significantly fewer layers, and performs adaptive multi-round iteration through recurrent latent reasoning via Hidden States, ultimately emitting refined, high-information-density tokens in a single burst.

In constructing the shared knowledge repository, we opted for a straightforward horizontal concatenation, as the FFN is conventionally regarded as a Key-Value Knowledge-Vector Pool, and this concatenation approach preserves the correspondence relationships among these knowledge vectors. Although such simple concatenation yields performance benefits, excessive parameter activation degrades the model's training and inference efficiency. Therefore, at larger parameter scales, we employ a block-wise partitioning technique analogous to MoE [63] to partition the FFN, with sparse activation during forward propagation.

Table 2: Step-aligned comparison between Intern-S2-Mobius-35B and Qwen3.5-35B on a linear-algebra multiple-choice question. Both models select the correct answer (Option C). Token counts are computed using the Qwen3.5-35B tokenizer. Mobius completes the same reasoning steps with fewer tokens, which mainly benefits from the model's elimination of repeated derivation and checks.

Reasoning Trace Comparison

Aligned Step	**Intern-S2-Mobius-35B**	**Qwen3.5-35B**
Task framing	**17 tokens.** "The user wants me to evaluate the truth value of two statements regarding linear algebra."	**20 tokens.** "The user wants me to evaluate two statements and determine their truth values from a list of options."
Statement 1	**178 tokens.** "Counter-example: Let $v_1 = (1, 0)$ and $v_2 = (2, 0)$. Then $2v_1 - v_2 = 0$, so they are linearly dependent. … Statement 1 is False."	**283 tokens.** "Consider the zero vector. Let $v_1 = (1, 0)$ and $v_2 = (0, 0)$. … Consider two parallel vectors. Let $v_1 = (1, 0)$ and $v_2 = (2, 0)$. … Statement 1 is **False**."
Statement 2	**142 tokens.** "A basis is a linearly independent spanning set. Since $\{v_1, \dots, v_k\}$ spans V and is linearly independent, it forms a basis for V. … $\dim(V) = k$."	**197 tokens.** "The set forms a basis for V if it spans V and is linearly independent. … Therefore, $\{v_1, \dots, v_k\}$ is a basis for V. … $\dim(V) = k$."
Option matching	**22 tokens.** "Statement 1 is False, Statement 2 is True. This matches Option C."	**122 tokens.** "The combination is False, True. This corresponds to Option C."
Repeated derivation and checks	—	**1,147 tokens.** "**Step-by-step derivation:** … Counterexample: Let $v_1 = (1, 0)$ and $v_2 = (2, 0)$. … **Final check:** … If $k = 0$? … Works. … Option C."
Visible final answer	**157 tokens.** "The vectors $(1, 0)$ and $(2, 0)$ are linearly dependent because one is a scalar multiple of the other. … This is the definition of a basis. … ANSWER: C"	**595 tokens.** "**Step 1: Evaluate Statement 1** … $2v_1 - 1v_2 = (0, 0)$. … Statement 1 is **False**. **Step 2: Evaluate Statement 2** … $\dim(V)$ is equal to k. … ANSWER: C"
Total	**516 tokens**	**2,364 tokens**

3. How about Mobius' performance?

To validate the performance of Mobius, we conducted both training-from-scratch (TFS) and continual pre-training (CPT) experiments. For the TFS experiments, we trained 7B-A1B MoE models — one with Mobius and one with Transformer — on 1TB tokens, and compared their MMLU scores. For the CPT experiments, we used Qwen3.5-35B-A3B as the starting checkpoint, continued pre-training on 1TB tokens, and subsequently performed supervised fine-tuning (SFT) and reinforcement learning (RL).

3.1. Mobius delivers markedly better data efficiency than Transformer.

As shown in Figure 2, Mobius achieves significantly higher MMLU scores than Transformer across all stages of training. Moreover, when using Transformer's score at 1TB tokens as the baseline, Mobius attains the same score with only 0.626× data — that is, Mobius exhibits 1.6× data efficiency of Transformer.

The mechanism by which Mobius achieves higher data efficiency remains to be fully explored. Nevertheless, we hypothesize that under the Transformer architecture, parameters across layers exhibit considerable redundancy—for instance, the same piece of knowledge may be redundantly stored in multiple layers. Upon switching to the knowledge-reasoning-decoupled Mobius architecture, the model can attain superior compression rates, thereby enabling it to acquire sufficient knowledge with substantially less training data.

3.2. Mobius matches Transformer-level reasoning with far greater inference efficiency.

Given the prohibitive cost of pre-training from scratch, to validate Mobius's capabilities at scale, we chose to continue training from an existing open-source base model. As shown in Table 1, starting from Qwen3.5 and switching to the Mobius architecture for continued pre-training not only preserves but actually enhances the model's overall capabilities.

More notably, switching to Mobius not only preserves fundamental reasoning capabilities but also improves end-to-end inference efficiency. As shown in Figure 3, aggregated across multiple evaluation benchmarks, Mobius achieves substantially higher request throughput than Transformer. We also examined the inference length of both models, visualized in Figure 4, which reveals that the improvement in Mobius's end-to-end inference efficiency primarily stems from shorter CoT lengths. When presented with identical problems, Mobius resolves them with markedly shorter reasoning chains, whereas Transformer requires longer CoT for deliberation.

Although the precise mechanism behind Mobius's shorter CoT reasoning remains not fully established, we hypothesize that this may be attributed to Mobius's native latent reasoning characteristic, which internalizes the deliberation process within the model and achieves more efficient reasoning through continuously differentiable optimization.

However, whether Mobius merely hacks the problem or genuinely provides a more efficient mode of reasoning remains to be determined. By comparing multiple cases, we find that the shortened CoT in Mobius stems from the latter. The case in Table 2 is from MMLU-Pro Mathematics and tests two basic linear-algebra statements. The first statement is false because two vectors in $\mathbb{R}^2$ need not be linearly independent. The second statement is true because a linearly independent set that spans V is a basis of V, so its k vectors imply $\dim(V) = k$. The correct answer is therefore Option C: False, True.

4. Mobius' relationship with mainstream research

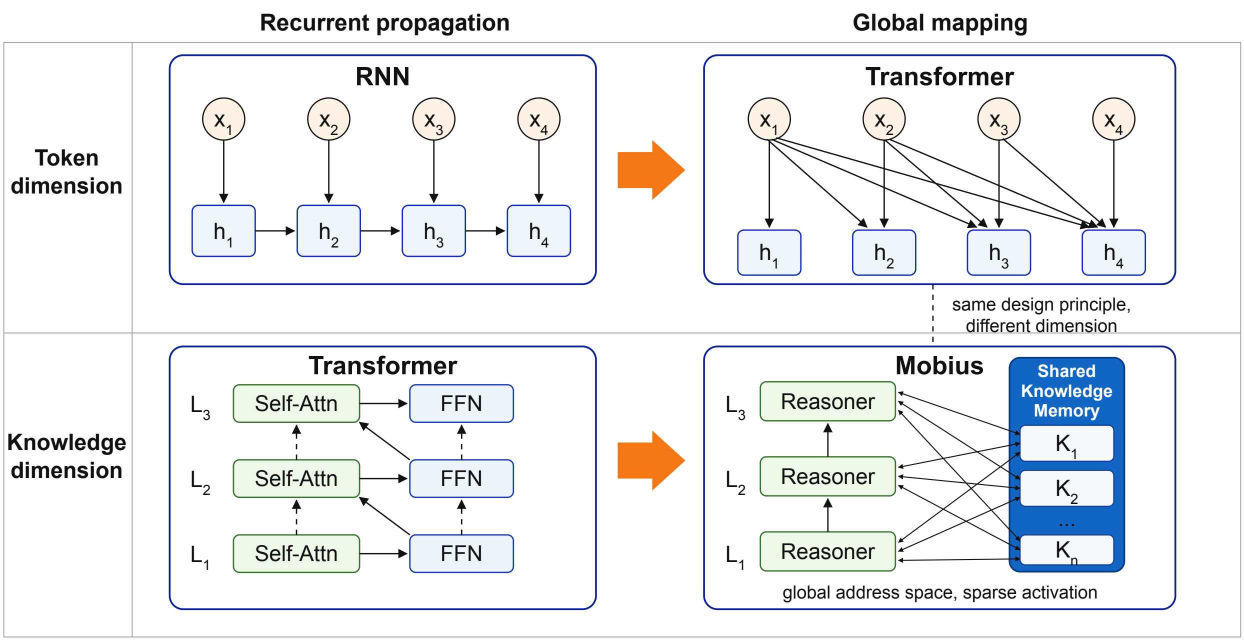

Figure 5: The comparison between RNN, Transformer, and Mobius.

4.1. Latent Reasoning

Chain-of-thought (CoT) externalizes reasoning into intermediate tokens [71], but incurs sequential decoding overhead. Latent reasoning instead performs computation in continuous states, exploring three directions: continuous thought, looped computation, and parallel refinement.

4.1.1. Continuous Thought

Continuous-thought approaches replace discrete reasoning tokens with differentiable representations passed between reasoning steps. COCONUT directly feeds the final hidden state back as the next input embedding, allowing intermediate computation to proceed without decoding each state into language [30]. CODI compresses explicit CoT supervision into continuous states through self-distillation [64], while SoftCoT generates instance-specific soft thoughts with a lightweight assistant and projects them into the representation space of a target language model [76]. These approaches demonstrate that continuous states can preserve useful reasoning information while shortening or eliminating explicit rationales.

These approaches expose a small set of latent states that subsequent tokens can attend to and reuse as shared context. Mobius further makes this pattern a native architectural capability: its recurrent states are refined against the shared Memory and jointly support multiple future tokens, yielding higher-density reasoning without verbose traces.

4.1.2. Looped Language Models

Looped language models increase effective depth by repeatedly applying shared computation blocks. Universal Transformers introduced recurrent refinement across depth with parameter sharing [12], while subsequent analyses established the computational expressivity of Looped Transformers [21]. More recent work connects effective depth directly to reasoning: looped models can emulate multi-step latent computation and approach much deeper non-looped models on reasoning tasks [60], while recurrent-depth pre-training enables test-time compute scaling by increasing the number of latent iterations without generating additional reasoning tokens [19].

Mobius also scales computation through latent recurrence, but performs each update over only a few layers while retaining access to the full shared Memory. This higher update frequency enables more iterative refinement and concentrates reasoning into higher-information-density latent states.

4.1.3. Additional Latent Computation Steps

Another line of work increases reasoning capacity by introducing extra latent computation steps before generation. Pause-token methods add learned placeholders to defer prediction and create additional computation space [24]. Hidden Decoding expands each token into multiple latent streams within a single forward pass, using intermediate key–value states as reusable computation traces [46]. Diffusion language models achieve similar effects through iterative refinement of continuous or masked representations [43, 55]. These methods shift computation from explicit CoT traces to latent trajectories.

Mobius extends this idea through recurrent latent refinement without explicit placeholders or diffusion steps. Its Reasoners repeatedly retrieve from a shared knowledge-vector Memory, refine high-density latent states, and decode multiple tokens in parallel.

4.2. Efficient Reasoning

Historically, the predominant paradigm for language model inference relied on autoregressive single-token generation from a single strong model, augmented with long chains of thought. To improve end-to-end inference efficiency, two approaches have gained increasing traction: Speculative Decoding, which enhances inference parallelism, and Concise CoT, which reduces the sequential length of reasoning chains.

4.2.1. Speculative Decoding

Speculative decoding accelerates autoregressive generation by using a cheap drafter to propose a block, or a tree, of future tokens and verifying these candidates in parallel with the target model [8, 42, 53, 68]. With rejection-sampling correction, this verification preserves the target model's output distribution. While classical methods employ a lightweight external draft model, later work develops internal drafters based on auxiliary decoding heads or predicted hidden features [2, 7, 44]. MTP is such an internal drafting mechanism rather than a decoding protocol itself: auxiliary heads sharing the target backbone predict multiple future tokens [22, 47, 51].

Whether employing a draft model or other strategies, Speculative Decoding traverses all knowledge only once before token prediction. In contrast, Mobius performs multiple rounds of knowledge traversal and multi-token iteration internally, yielding hidden states of higher information density prior to final decoding, with native support for multi-token prediction.

4.2.2. Concise Chain of Thought

Long chain-of-thought (CoT) reasoning can improve performance on challenging tasks, but its autoregressive generation incurs substantial inference cost and latency. Recent work therefore seeks concise CoT through several complementary approaches: supervised methods learn from pruned or paired long–short trajectories, or internalize explicit rationales into implicit or dense representations [9, 13, 14, 36, 73, 79]; RL-based methods shape reasoning length through reward design, including controllable compression and length-aware objectives [11, 18, 50, 65]. Although RL can elicit advanced behaviors such as self-correction, verification, and backtracking [26, 62], outcome-only rewards often favor longer trajectories because additional tokens enable further exploration and error correction [77]. The central challenge is thus to reduce CoT verbosity without removing the deliberation needed for accurate and robust reasoning.

Conventional length regularization may shorten CoT by curtailing deliberation, potentially harming accuracy and robustness. In contrast, Mobius yields far more refined and shorter responses through several rounds of latent thought iteration.

4.3. Architecture Design

Mobius redesigns the internal information flow of Transformer architectures by decoupling knowledge storage from reasoning computation. In this section, we discuss its relationship with two major architectural directions: attention mechanisms and residual connections.

4.3.1. Attention Mechanism

Self-attention is the core component of Transformer architectures, enabling models to capture global dependencies through pairwise interactions among tokens [69]. However, the quadratic complexity of full attention has become a major bottleneck for scaling Transformer models to long-context scenarios. To address this issue, extensive studies have explored more efficient attention mechanisms. Sparse attention reduces unnecessary token interactions by restricting attention patterns, while linear attention reformulates the attention computation to reduce complexity through alternative kernelization or factorization strategies. More recently, state-space models and gated state-space variants, such as Mamba and Gated Delta Networks, replace explicit attention computation with recurrent state updates, providing efficient alternatives for long-sequence modeling [25, 78]. These approaches mainly improve efficiency by modifying the computation form or reducing the cost of information interaction.

Rather than reducing attention complexity directly, we aim to increase the information density and value extracted per attention operation. Specifically, by decoupling knowledge from reasoning and constructing a globally shared knowledge repository, Mobius incur higher retrieval costs but provide attention with a more flexible selection space and higher-quality inputs, thereby completing reasoning with a more compact computation trajectory.

4.3.2. Residual Connection

Residual connections are fundamental components for scaling deep neural networks, enabling stable optimization and effective information propagation through shortcut pathways [31]. Subsequent studies have explored more flexible connection patterns to improve information transmission across layers. Highway Networks introduce learnable gates to adaptively control information flow [66], while DenseNet increases connection density by allowing each layer to directly access previous representations [35]. More recently, Hyper-Connections and Attention Residuals further investigate how to enhance residual pathways by increasing connection capacity or dynamically aggregating previous-layer representations [39, 80]. Despite these improvements, existing residual enhancement methods mainly focus on strengthening forward information propagation, where information flows from earlier layers to later layers.

Mobius extends residual design from forward information propagation toward bidirectional knowledge access, instead of explicitly expanding residual pathways. Since all reasoning stages can access the same knowledge repository, deeper Reasoners can retrieve knowledge beyond their local layer hierarchy, enabling more flexible information transmission while maintaining an efficient computation structure.

5. Mobius' potential on several highlight topics

5.1. Self-Evolving

Although models and agents have grown increasingly capable, how to enable them to continuously absorb new knowledge and skills from external sources remains an open question [40, 56, 59, 72].

Our position is that, **the current Transformer architecture does not satisfy the prerequisites for self-evolution.** Because knowledge and reasoning are tightly coupled in Transformer, such models can only acquire new skills through end-to-end training, which inevitably alters both knowledge and reasoning capabilities simultaneously, thereby causing catastrophic forgetting of previously learned skills. An architecture suitable for self-evolution should exhibit a certain degree of knowledge-reasoning decoupling, as this would support unbounded and efficient expansion of knowledge storage, and enable the model's reasoning capabilities to generalize across multiple domains. The Mobius architecture we propose possesses preliminary knowledge-reasoning separation characteristics, demonstrating greater potential for continual learning compared to Transformer.

Nevertheless, whether Mobius can perform better in real-world self-evolution scenarios still requires the joint design of foundation models, agentic systems, and environmental feedback, and remains to be validated in future work.

5.2. World Model

World models are emerging as the next frontier beyond language models, yet how to enable models to perform understanding and reasoning over continuous-space inputs remains an open question [6, 27–29, 33, 61].

Our position is that, **the current Transformer architecture does not satisfy the prerequisites for world models.** Models built primarily upon Transformer have been demonstrated to model discrete modalities (such as language) at the terabyte parameter scale, yet perfect modeling of continuous space with Transformer would likely require parameter scales on the order of petabytes. The inference cost of terabyte-scale language models already strains existing hardware to its limits; the inference cost of a petabyte-scale world model would be astronomical. The Mobius architecture we propose natively supports latent reasoning, offering a stronger prior for continuous-space modeling and demonstrating greater potential than Transformer.

Nevertheless, the latent reasoning prior introduced by Mobius may prove far from sufficient; how to better reengineer the attention mechanism may be of even greater criticality.

5.3. Scientific Discovery

Contemporary AI has achieved remarkable proficiency in code and mathematics, and recent scientific foundation models excel at procedural problem-solving [5, 58]. Nevertheless, enabling models to formulate disruptive scientific hypotheses and ideas remains an open challenge [23, 48, 70].

Our position is that, **the current paradigm of Transformer architecture combined with long chains of thought does not satisfy the prerequisites for scientific discovery.** Under the present paradigm, models excel at procedural problem-solving; however, scientific discovery demands not only procedural competence but also scientific intuition and the composition and generalization of knowledge. The Mobius architecture addresses this on two fronts: on one hand, it internalizes deliberation into a continuously differentiable latent space [30, 64], affording the model the opportunity to develop more powerful intuition; on the other hand, it flattens all knowledge within the model, enabling a greater volume of knowledge to interact simultaneously and substantially increasing the possibility of compositional generalization across knowledge domains.

Nevertheless, whether Mobius can perform better in real-world scientific discovery scenarios likely requires the joint optimization of data, infrastructure, training algorithms, and optimization algorithms, and remains to be validated in future work.

5.4. Hardware-Software Co-Design

Models are being scaled to ever-larger parameter counts, demonstrating that greater scale does yield emergent intelligence, while simultaneously pushing the limits of physical hardware [6, 33, 37, 41, 57].

Our position is that, **the current combination of Transformer architecture and existing hardware systems does not satisfy the prerequisites for further scaling.** This is because the Transformer architecture commingles knowledge and reasoning within the same parameter set, necessitating the loading of nearly all parameters into GPU memory at deployment. The Mobius architecture we propose exhibits knowledge-reasoning separation, opening the possibility of stably retaining only reasoning-dedicated parameters in GPU memory while storing knowledge parameters predominantly on SSD, with high-priority knowledge retrieved and loaded into memory on demand.

In the long term, Mobius is more amenable to hardware-software co-design than Transformer. However, substantial effort remains before this vision can be realized.

References

[1] Zeyuan Allen-Zhu. ICML 2024 Tutorial: Physics of Language Models, July 2024. Project page: https://physics.allen-zhu.com/. C

[2] Zachary Ankner, Rishab Parthasarathy, Aniruddha Nrusimha, Christopher Rinard, Jonathan Ragan-Kelley, and William Brandon. Hydra: Sequentially-dependent draft heads for medusa decoding. *arXiv preprint arXiv:2402.05109*, 2024. 4.2.1

[3] Simran Arora, Sabri Eyuboglu, Michael Zhang, Aman Timalsina, Silas Alberti, Dylan Zinsley, James Zou, Atri Rudra, and Christopher Ré. Simple linear attention language models balance the recall-throughput tradeoff, 2025. 1

[4] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. Is space-time attention all you need for video understanding? In *ICML*, 2021. 1

[5] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624:570–578, 2023. 5.3

[6] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. In *NeurIPS*, 2020. 5.2, 5.4

[7] Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, Deming Chen, and Tri Dao. Medusa: Simple llm inference acceleration framework with multiple decoding heads, 2024. 4.2.1

[8] Charlie Chen, Sebastian Borgeaud, Geoffrey Irving, Jean-Baptiste Lespiau, Laurent Sifre, and John Jumper. Accelerating large language model decoding with speculative sampling, 2023. 4.2.1

[9] Jeffrey Cheng and Benjamin Van Durme. Compressed chain of thought: Efficient reasoning through dense representations. *arXiv preprint arXiv:2412.13171*, 2024. 2.2, 4.2.2

[10] Krzysztof Choromanski, Valerii Likhosherstov, Xingyou Song, Richard Davis, Kyle Sarlos, and Adrian Weller. Rethinking attention with performers. In *ICLR*, 2021. 1

[11] Muzhi Dai, Shixuan Liu, and Qingyi Si. Stable reinforcement learning for efficient reasoning, 2025. 4.2.2

[12] Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Lukasz Kaiser. Universal transformers. In *ICLR*. OpenReview.net, 2019. 4.1.2

[13] Yuntian Deng, Yejin Choi, and Stuart Shieber. From explicit cot to implicit cot: Learning to internalize cot step by step. *arXiv preprint arXiv:2405.14838*, 2024. 4.2.2

[14] Yuntian Deng, Kiran Prasad, Roland Fernandez, Paul Smolensky, Vishrav Chaudhary, and Stuart Shieber. Implicit chain of thought reasoning via knowledge distillation, 2023. 4.2.2

[15] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In Jill Burstein, Christy Doran, and Thamar Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics. 1

[16] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale, 2021. 1

[17] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. https://transformer-circuits.pub/2021/framework/index.html. 2

[18] Mehdi Fatemi, Banafsheh Rafiee, Mingjie Tang, and Kartik Talamadupula. Concise reasoning via reinforcement learning, 2025. 4.2.2

[19] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. In D. Belgrave, C. Zhang, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, and N. Chen, editors, *Advances in Neural Information Processing Systems*, volume 38, pages 41340–41391. Curran Associates, Inc., 2025. 4.1.2

[20] Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer feed-forward layers are key-value memories. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih, editors, *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 5484–5495, Online and Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. 2

[21] Angeliki Giannou, Shashank Rajput, Jy yong Sohn, Kangwook Lee, Jason D. Lee, and Dimitris Papailiopoulos. Looped transformers as programmable computers. In *ICML*, pages 11398–11442, 2023. 2.2, 4.1.2

[22] Fabian Gloeckle, Badr Youbi Idrissi, Baptiste Rozière, David Lopez-Paz, and Gabriel Synnaeve. Better & faster large language models via multi-token prediction. In *Proceedings of the 41st International Conference on Machine Learning*, pages 15706–15734, 2024. 4.2.1

[23] Juraj Gottweis, Wei-Hung Weng, Alexander Daryin, Tao Tu, et al. Accelerating scientific discovery with co-scientist. *Nature*, 655(8122):487–496, 2026. 5.3

[24] Sachin Goyal, Ziwei Ji, Ankit Singh Rawat, Aditya Krishna Menon, Sanjiv Kumar, and Vaishnavh Nagarajan. Think before you speak: Training language models with pause tokens. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, editors, *International Conference on Learning Representations*, volume 2024, pages 27896–27923, 2024. 4.1.3

[25] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *First Conference on Language Modeling*, 2024. 1, 4.3.1

[26] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, Aixin Liu, Bing Xue, Bingxuan Wang, Bochao Wu, Bei Feng, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chong Ruan, Damai Dai, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Hanwei Xu, Honghui Ding, Huazuo Gao, Hui Qu, Hui Li, Jianzhong Guo, Jiashi Li, Jingchang Chen, Jingyang Yuan, Jinhao Tu, Junjie Qiu, Junlong Li, J. L. Cai, Jiaqi Ni, Jian Liang, Jin Chen, Kai Dong, Kai Hu, Kaichao You, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Liang Zhao, Litong Wang, Liyue Zhang, Lei Xu, Leyi Xia, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Mingxu Zhou, Meng Li, Miaojun Wang, Mingming Li, Ning Tian, Panpan Huang, Peng Zhang, Qiancheng Wang, Qinyu Chen, Qiushi Du, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, R. J. Chen, R. L. Jin, Ruyi Chen, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shengfeng Ye, Shiyu Wang, Shuiping Yu, Shunfeng Zhou, Shuting Pan, S. S. Li, Shuang Zhou, Shaoqing Wu, Tao Yun, Tian Pei, Tianyu Sun, T. Wang, Wangding Zeng, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, W. L. Xiao, Wei An, Xiaodong Liu, Xiaohan Wang, Xiaokang Chen, Xiaotao Nie, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, X. Q. Li, Xiangyue Jin, Xiaojin Shen, Xiaosha Chen, Xiaowen Sun, Xiaoxiang Wang, Xinnan Song, Xinyi Zhou, Xianzu Wang, Xinxia Shan, Y. K. Li, Y. Q. Wang, Y. X. Wei, Yang Zhang, Yanhong Xu, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Wang, Yi Yu, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He,

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[27] David Ha and Jürgen Schmidhuber. World models. In *Advances in Neural Information Processing Systems*, volume 31, 2018. 5.2

[28] Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning*, pages 2555–2565. PMLR, 2019.

[29] Danijar Hafner, Jürgen Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023. 5.2

[30] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space. *arXiv preprint arXiv:2412.06769*, 2024. 4.1.1, 5.3

[31] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016. 2, 2.1, 4.3.2

[32] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Identity mappings in deep residual networks. In *European conference on computer vision*, pages 630–645. Springer, 2016. 2.1

[33] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Oriol Vinyals, Jack W. Rae, and Laurent Sifre. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 30016–30030. Curran Associates, Inc., 2022. 1, 1, 5.2, 5.4

[34] Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. Densely connected convolutional networks. In *CVPR*, pages 4700–4708, 2017. 2, 2.1

[35] Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 4700–4708, 2017. 4.3.2

[36] Yu Kang, Xianghui Sun, Liangyu Chen, and Wei Zou. C3ot: Generating shorter chain-of-thought without compromising effectiveness. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 24312–24320, 2025. 4.2.2

[37] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020. 1, 1, 5.4

[38] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are rnns: Fast autoregressive transformers with linear attention. In *ICML*, pages 5156–5165, 2020. 1

[39] Kimi Team. Attention residuals. *arXiv preprint arXiv:2603.15031*, 2026. 2.1, 4.3.2

[40] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017. 5.1

[41] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with pagedattention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, pages 611–626. ACM, 2023. 5.4

[42] Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *International Conference on Machine Learning*, pages 19274–19286. PMLR, 2023. 4.2.1

[43] Xiang Li, John Thickstun, Ishaan Gulrajani, Percy S Liang, and Tatsunori B Hashimoto. Diffusion-lm improves controllable text generation. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors, *Advances in Neural Information Processing Systems*, volume 35, pages 4328–4343. Curran Associates, Inc., 2022. 2.2, 4.1.3

[44] Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. Eagle: speculative sampling requires rethinking feature uncertainty. In *Proceedings of the 41st International Conference on Machine Learning*, pages 28935–28948, 2024. 4.2.1

[45] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step, 2023. 1, 2.2

[46] Aiwei Liu, Cheng Shi, Chuhan Wu, Ci Lei, Di Lu, Donald He, Fan Zhang, Fanhao Kong, Feifei Zhang, Guan Wang, et al. Hidden decoding at scale: Latent computation scaling for large language models. *arXiv preprint arXiv:2607.08186*, 2026. 4.1.3

[47] Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, et al. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*, 2024. 4.2.1

[48] Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The ai scientist: Towards fully automated open-ended scientific discovery, 2024. 5.3

[49] Wenjie Ma, Jingxuan He, Charlie Snell, Tyler Griggs, Sewon Min, and Matei Zaharia. Reasoning models can be effective without thinking, 2025. 1

[50] Xinyin Ma, Guangnian Wan, Runpeng Yu, Gongfan Fang, and Xinchao Wang. Cot-valve: Length-compressible chain-of-thought tuning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6025–6035, 2025. 4.2.2

[51] Somesh Mehra, Javier Alonso Garcia, and Lukas Mauch. On multi-token prediction for efficient llm inference, 2025. 4.2.1

[52] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors, *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372. Curran Associates, Inc., 2022. 2

[53] Xupeng Miao, Gabriele Oliaro, Zhihao Zhang, Xinhao Cheng, Zeyu Wang, Zhengxin Zhang, Rae Ying Yee Wong, Alan Zhu, Lijie Yang, Xiaoxiang Shi, et al. Specinfer: Accelerating large language model serving with tree-based speculative inference and verification. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 3*, pages 932–949, 2024. 4.2.1

[54] Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori Hashimoto. s1: Simple test-time scaling. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 20275–20321, Suzhou, China, November 2025. Association for Computational Linguistics. 1

[55] Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan LI. Large language diffusion models. In D. Belgrave, C. Zhang, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, and N. Chen, editors, *Advances in Neural Information Processing Systems*, volume 38, pages 50608–50646. Curran Associates, Inc., 2025. 2.2, 4.1.3

[56] German I. Parisi, Ronald Kemker, Jose L. Part, Christopher Kanan, and Stefan Wermter. Continual lifelong learning with neural networks: A review. *Neural Networks*, 113:54–71, 2019. 5.1

[57] Samyam Rajbhandari, Jeff Rasley, Olatunji Ruwase, and Yuxiong He. Zero: Memory optimizations toward training trillion parameter models. In *SC20: International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 1–16. IEEE, 2020. 5.4

[58] Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M. Pawan Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan Ellenberg, et al. Funsearch: Making new discoveries in mathematical sciences using large language models. *Nature*, 625:468–475, 2024. 5.3

[59] Adam Santoro, Sergey Bartunov, Matthew M. Botvinick, Daan Wierstra, and Timothy P. Lillicrap. One-shot learning with memory-augmented neural networks. *CoRR*, abs/1605.06065, 2016. 5.1

[60] Nikunj Saunshi, Nishanth Dikkala, Zhiyuan Li, Sanjiv Kumar, and Sashank J. Reddi. Reasoning with latent thoughts: On the power of looped transformers. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu, editors, *International Conference on Learning Representations*, volume 2025, pages 14855–14881, 2025. 4.1.2

[61] Julian Schrittwieser, Ioannis Antonoglou, Thomas Hubert, Karen Simonyan, Laurent Sifre, Simon Schmitt, Arthur Guez, Edward Lockhart, Demis Hassabis, David Silver, et al. Mastering atari, go, chess and shogi by planning with a learned model. *Nature*, 588:604–609, 2020. 5.2

[62] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. 4.2.2

[63] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Sparsely-gated mixture-of-experts layers. In *ICLR*, 2017. 2.3

[64] Zhenyi Shen, Hanqi Yan, Linhai Zhang, Zhanghao Hu, Yali Du, and Yulan He. CODI: Compressing chain-of-thought into continuous space via self-distillation. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 677–693, Suzhou, China, November 2025. Association for Computational Linguistics. 4.1.1, 5.3

[65] Mingyang Song and Mao Zheng. Walk before you run! concise llm reasoning via reinforcement learning, 2025. 4.2.2

[66] Rupesh K. Srivastava, Klaus Greff, and Jürgen Schmidhuber. Training very deep networks. In *Advances in Neural Information Processing Systems*, pages 2377–2385, 2015. 4.3.2

[67] Rupesh K. Srivastava, Klaus Greff, and Jürgen Schmidhuber. Highway networks. In *ICML Deep Learning Workshop*, 2015. 2

[68] Mitchell Stern, Noam Shazeer, and Jakob Uszkoreit. Blockwise parallel decoding for deep autoregressive models. *Advances in Neural Information Processing Systems*, 31, 2018. 4.2.1

[69] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Ł ukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. 1, 2, 4.3.1

[70] Yuru Wang, Lejun Cheng, Yuxin Zuo, Sihang Zeng, Bingxiang He, Che Jiang, Junlin Yang, Yuchong Wang, Kaikai Zhao, Weifeng Huang, Kai Tian, Zhenzhao Yuan, Jincheng Zhong, Weizhi Wang, Ning Ding, Bowen Zhou, and Kaiyan Zhang. Naturebench: Can coding agents match the published sota of nature-family papers?, 2026. 5.3

[71] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS*, 2022. 1, 2, 2.2, 4.1

[72] Jason Weston, Sumit Chopra, and Antoine Bordes. Memory networks. In *ICLR*, 2015. 5.1

[73] Heming Xia, Chak Tou Leong, Wenjie Wang, Yongqi Li, and Wenjie Li. Tokenskip: Controllable chain-of-thought compression in llms. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 3351–3363, 2025. 4.2.2

[74] Zhenda Xie, Yixuan Wei, Huanqi Cao, Chenggang Zhao, Chengqi Deng, Jiashi Li, Damai Dai, Huazuo Gao, Jiang Chang, Kuai Yu, Liang Zhao, Shangyan Zhou, Zhean Xu, Zhengyan Zhang, Wangding Zeng, Shengding Hu, Yuqing Wang, Jingyang Yuan, Lean Wang, and Wenfeng Liang. mhc: Manifold-constrained hyper-connections, 2026. 2.1

[75] Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tieyan Liu. On layer normalization in the transformer architecture. In *International conference on machine learning*, pages 10524–10533. PMLR, 2020. 2.1

[76] Yige Xu, Xu Guo, Zhiwei Zeng, and Chunyan Miao. SoftCoT: Soft chain-of-thought for efficient reasoning with LLMs. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 23336–23351, Vienna, Austria, July 2025. Association for Computational Linguistics. 4.1.1

[77] Shiming Yang, Yuxuan Tong, Xinyao Niu, Graham Neubig, and Xiang Yue. Demystifying long chain-of-thought reasoning. In *International Conference on Machine Learning*, pages 71177–71209. PMLR, 2025. 4.2.2

[78] Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving mamba2 with delta rule. In *The Thirteenth International Conference on Learning Representations*, 2025. 1, 4.3.1

[79] Ping Yu, Jing Xu, Jason Weston, and Ilia Kulikov. Distilling system 2 into system 1, 2024. 4.2.2

[80] Defa Zhu, Hongzhi Huang, Zihao Huang, Yutao Zeng, Yunyao Mao, Banggu Wu, Qiyang Min, and Xun Zhou. Hyper-connections. In *International Conference on Learning Representations*, 2025. 2.1, 4.3.2

[81] Yicheng Zou et al. Intern-s1-pro: Scientific multimodal foundation model at trillion scale, 2026. 1

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B. Expert Activation Patterns

We further compare expert-activation pattern under two training recipes introduced in 3. As shown in 6, the Mobius-7B model trained from scratch (left) exhibits a comparatively uniform activation distribution across reasoning layers. Intern-S2-Mobius-35B (right), continually pre-trained from Qwen3.5-35B, activates experts over a wider range while retaining a pronounced block-diagonal pattern. The latter suggests that expert routing remains influenced by the layer-specific organization of the source checkpoint.

These observations suggest that continual pre-training can broaden expert utilization without fully removing the routing prior induced by architectural conversion. Training Mobius from scratch may therefore permit more flexible access to the shared expert pool; whether this translates into stronger model capability requires controlled evaluation at comparable scale.

C. Toy Task of Compositional Generalization

We select a Compositional Generalization task from Physics of LLM [1]. On this task, we compared Transformer against our yet-to-be-released, newer Mobius architecture. As shown in Fig 7, Mobius achieved substantially better convergence efficiency and final scores. This compositional generalization task uses single-hop knowledge of 500 entities and two-hop knowledge of 400 entities as the training set, and two-hop knowledge of the remaining 100 entities as the test set. By comparing test-set scores, we examine whether the model can genuinely learn the connections between different knowledge pieces and achieve compositional generalization.

D. Case Study: Chain-of-Thought Reasoning

We further present an example from the field of biology in 3. In this case, Mobius also delivers a mush shorter reasoning trace than Transformer.

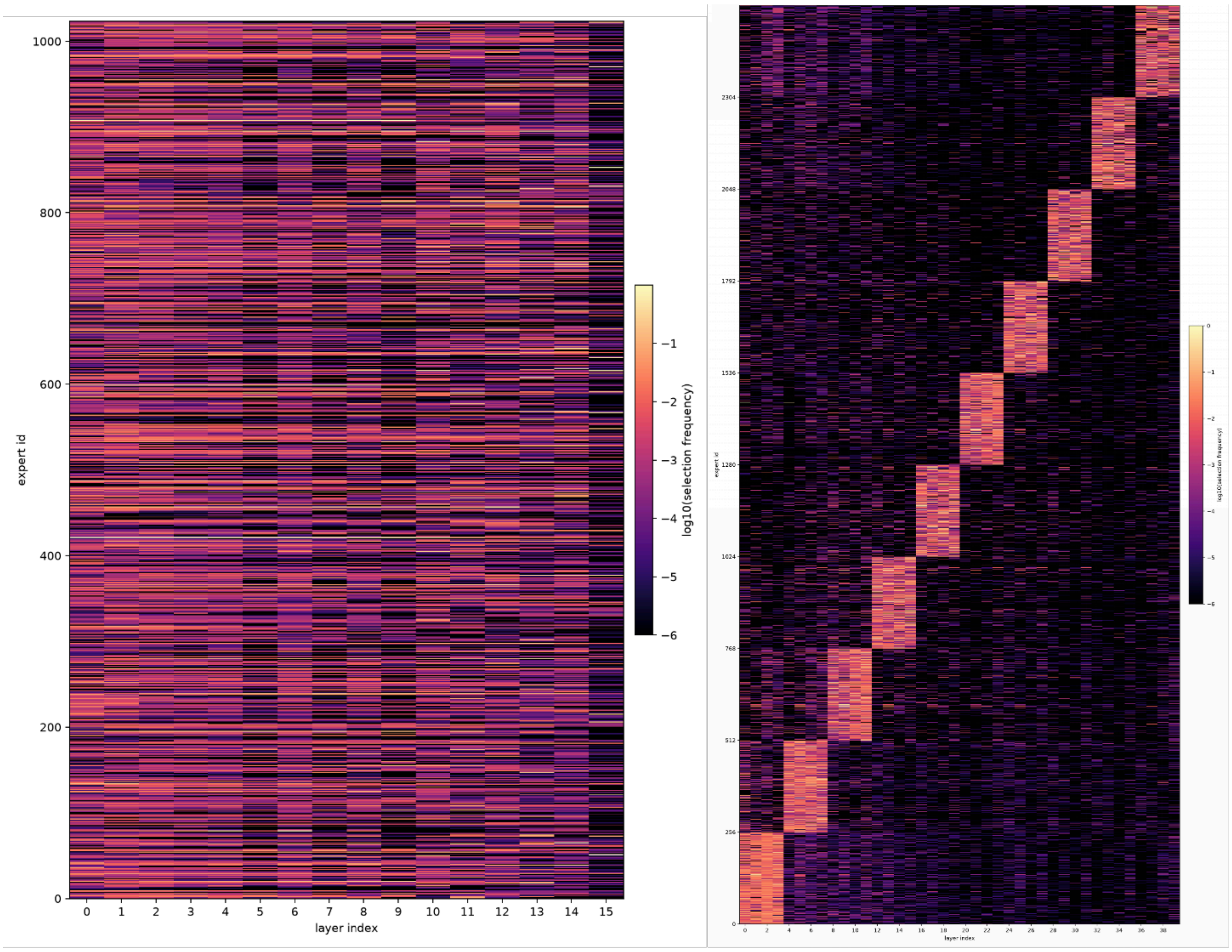
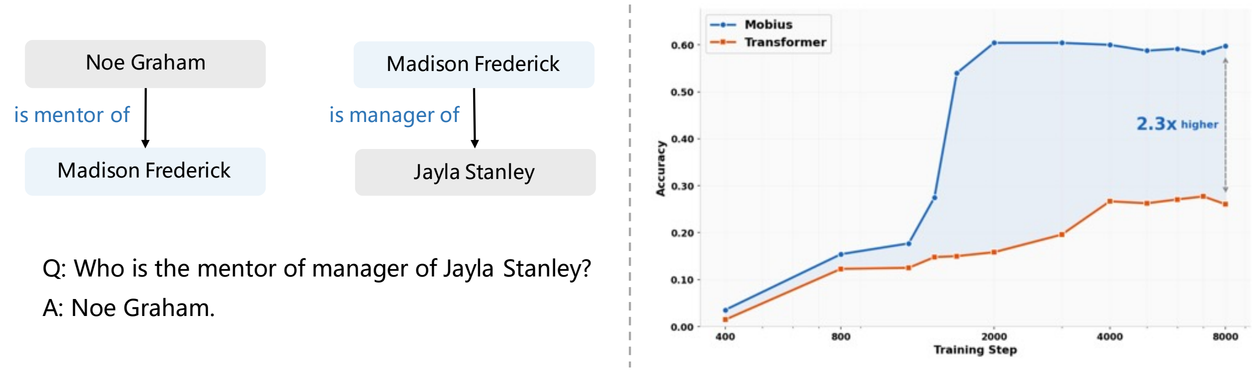


Figure 6: Expert-Activation pattern across layers under two training recipes. The left panel shows Mobius-7B trained from scratch; the right panel shows Intern-S2-Mobius-35B obtained by architecture conversion and continual pre-training. Rows denote expert IDs, columns denote layer indices, and color represents the base-10 logarithm of selection frequency. Expert IDs in the right panel are reordered to reveal the inherited routing structure.

E. Layerwise Analysis of Latent Reasoning

We use a layerwise prediction lens to examine how token predictions evolve during latent computation. Given the same teacher-forced context, we decode the hidden state at each layer for the standard next-token prediction ($t+1$) and four subsequent MTP positions ($t+2$ to $t+5$). As shown in Figure 8, Mobius exhibits more interpretable, target-aligned predictions in its intermediate layers than the baseline. Its layerwise predictions follow a more coherent semantic trajectory, suggesting that task-relevant information is concentrated into a compact internal representation rather than dispersed across competing token candidates. Further latent iterations progressively refine this representation and improve predictions at subsequent positions, ultimately yielding a fully accepted five-token draft. In contrast, the baseline exhibits less stable intermediate predictions, and its draft is rejected at the third predicted position, leaving only a two-token accepted prefix. This example suggests that Mobius may form more compact internal representations and refine them iteratively in latent space, potentially benefiting multi-token prediction.



(a) Task Description (a) Performance

Figure 7: The Compositional Generalization task selected from Physics of LLM. A future-version of Mobius(unreleased) enjoys faster convergence, higher final scores than Transformer. The train set contains 500-entity single-hop + 400-entity two-hop, while the test set contains 100-entity two-hop holdout. This task evaluates whether models learn cross-knowledge connections, not mere memorization.

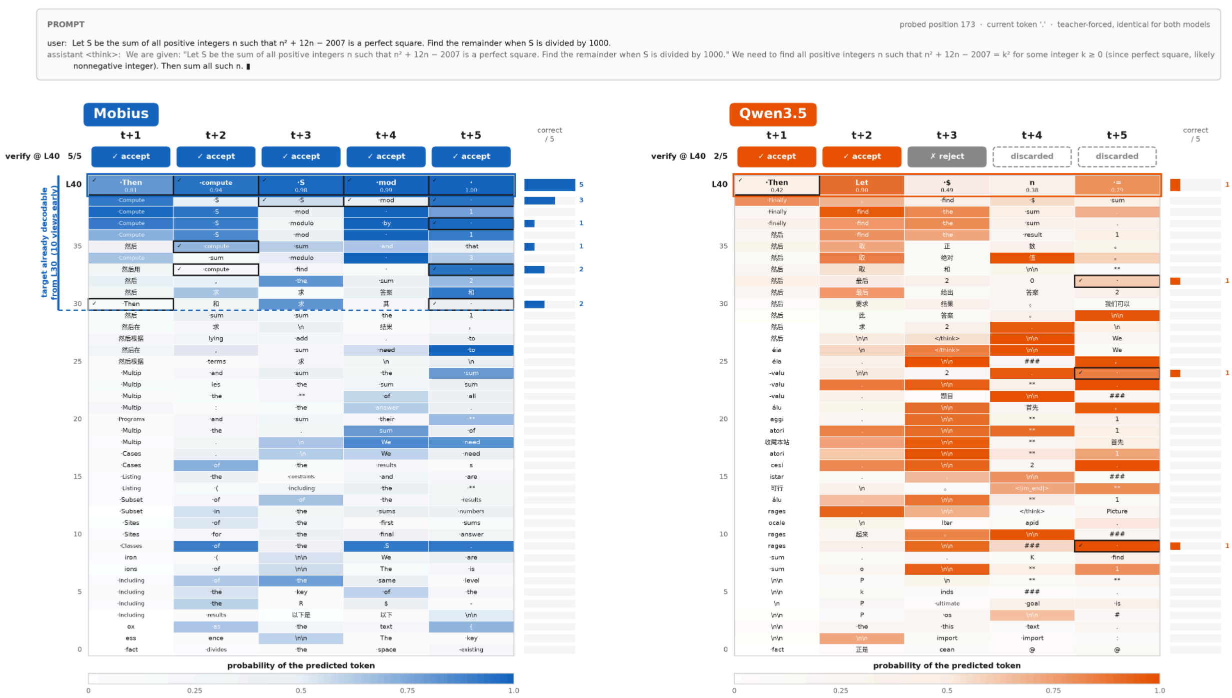


Figure 8: Layerwise prediction lens for Mobius and the Qwen3.5 baseline under an identical teacher-forced context. The $t+1$ column represents standard next-token prediction, while $t+2$ through $t+5$ are subsequent MTP predictions. Each cell shows the token predicted from a given hidden-state view; color intensity denotes its probability, and a black outline marks agreement with the target continuation. Mobius exhibits more target-aligned intermediate predictions and produces a five-token draft accepted in full, while the baseline produces only a two-token accepted prefix.

Table 3: Step-aligned comparison between Intern-Spin-35B and Qwen3.5-35B on a photosynthesis multiple-choice question. Both models select the correct answer (Option D). Token counts are computed using the Qwen3.5-35B tokenizer.

Reasoning Trace Comparison

Aligned Step	**Intern-Spin-35B**	**Qwen3.5-35B**
Task framing	**36 tokens.** “The user is asking a question about the relationship between the light reactions and the Calvin cycle in photosynthesis.”	**35 tokens.** “The user wants me to answer … **Question:** The light reactions … supply the Calvin cycle with which of the following?”
Core biological mechanism	**108 tokens.** “Light-dependent reactions … producing oxygen, ATP, and NADPH.” “Calvin cycle … uses ATP and NADPH … to fix CO_2 into sugars.”	**276 tokens.** “Light-dependent reactions … Outputs: Oxygen (O_2), ATP, NADPH. … Calvin Cycle … Inputs: CO_2, ATP, NADPH.” “… reduction requires ATP and NADPH …”
Option evaluation	**240 tokens.** “D: ATP provides energy, NADPH provides high-energy electrons …” “H: ATP and NADPH are produced in light reactions, not broken down there …”	**543 tokens.** “D … Correct. The light reactions generate ATP (energy) and NADPH (reducing power/electrons). …” “H … ATP and NADPH are produced in the light reactions and consumed … in the Calvin cycle.”
Repeated derivation	—	**205 tokens.** “**Conclusion:** Option D accurately describes the relationship … ATP and NADPH … are used in the Calvin cycle to fix carbon.” “**Step-by-step derivation:** … Light Reactions produce ATP and NADPH → Calvin Cycle consumes them. …”
Repeated option check	—	**143 tokens.** “**Final check:** … Option A: CO_2 source is wrong. … Option D: ATP/NADPH transfer is correct. …”
Reasoning closure	**13 tokens.** “Conclusion: Option D is the correct answer.”	**8 tokens.** “The correct option is D.”
Visible final answer	**75 tokens.** “The light reactions … stored in ATP and NADPH. These molecules are then used in the Calvin cycle …” “ANSWER: D”	**439 tokens.** “The process of photosynthesis is divided into two main stages … ATP and NADPH … are transported to the Calvin cycle.” “… D: This states ATP and NADPH are supplied. This is accurate. …” “ANSWER: D”
Total	**472 tokens**	**1,649 tokens**

Note: The token counts are sequential marginal counts over each raw prediction; therefore, they sum exactly to the total output length. The displayed excerpts are representative original text from the corresponding reasoning segment. Ellipses indicate omitted original text.